WITHDRAWN (2026-05-17)

*The main theorem of this note is **false**. At $t = 5$, a planar graph constructed by Voigt (1993) has $\chi_\ell(G) = 5$ (every planar graph is 5-choosable by Thomassen 1994; Voigt's example is not 4-choosable) and $\mathrm{cr}(G) = 0 < 1 = \mathrm{cr}(K_5)$. This directly falsifies "$\chi_\ell(G) \geq t$ and $t \leq 18 \Rightarrow \mathrm{cr}(G) \geq \mathrm{cr}(K_t)$" at $t = 5$.*

The structural cause is that the Albertson–Cranston–Fox / Barát–Tóth / Ackerman chain does not *use the chromatic-number hypothesis only through Dirac's $\delta \geq t - 1$ bound. Ackerman's argument (`arXiv:1509.01932`, Section 3.1) uses the minimum edge-count function $f_r(n)$ for r-critical graphs, i.e. sharper critical-graph machinery (e.g. Kostochka–Yancey-style bounds). The list-critical analogue (Krivelevich 1997 and follow-ups) gives a strictly weaker edge floor, so the "lifts for free" claim of this note is unjustified. Beyond the $t = 5$ counterexample, a more careful list-coloring analysis would also have to show that the weaker list-critical edge floor still suffices to drive the ACF/BT/Ackerman vertex bound through; we did not do this and likely it does not.*

The companion combined paper `D18_combined_observations/two_structural_observations.tex` that wraps this note as "Observation 1" is also withdrawn. This file is preserved for historical record only.

List-coloring Albertson up to chromatic number 18

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Abstract

Albertson's conjecture asserts that every graph G with chromatic number at least t satisfies $\mathrm{cr}(G) \geq \mathrm{cr}(K_t)$. We observe that the unconditional Albertson chain through chromatic number 18 – due to Albertson–Cranston–Fox [3], Barát–Tóth [4], and Ackerman [1] – depends on its χ-hypothesis only through the Dirac minimum-degree bound and the resulting edge floor on a colour-critical graph. Both inputs admit a clean lift to list-colouring via the list versions of Dirac's bound and Brooks' theorem (Borodin [6]; Erdős–Rubin–Taylor [13]). Assembling these observations gives the following theorem.

Every graph G with $\chi_\ell(G) \geq t$ and $t \leq 18$ satisfies $\mathrm{cr}(G) \geq \mathrm{cr}(K_t)$.

Because $\chi_\ell(G) \geq \chi(G)$ for every graph G, this is strictly stronger than ordinary Albertson at $t \leq 18$. The same argument yields a corresponding statement for DP-colouring. We also explain why the same lift currently *stops* at $t = 18$: the recent extension of ordinary Albertson to $t \leq 24$ by Cranston [9] uses the Fox–Pach–Suk chromatic-index lemma [14] at the sharp constant 9/16, whose list-edge-colouring analogue at this constant is not currently in the literature. The best published list-edge-colouring bound (Borodin–Kostochka–Woodall [7]) is roughly a factor 3 weaker, which is enough to collapse the Fox–Pach–Suk vertex bound below the Ackerman threshold.

1 Introduction

Albertson's conjecture [2] is the natural common strengthening of the Four Colour Theorem and Guy's formula for $\mathrm{cr}(K_n)$:

Conjecture 1 (Albertson)**.** Every graph G with $\chi(G) \geq t$ satisfies $\mathrm{cr}(G) \geq \mathrm{cr}(K_t)$.

The conjecture is currently known unconditionally for $t \leq 24$ through a chain of partial results that all proceed by analysing a minimum counterexample (MCE). Each step combines two ingredients:

1. an *edge floor* on critical graphs – $|E(G)| \geq \frac{k-1}{2}|V(G)|$ via Dirac's theorem on critical graphs [11];
2. a *Crossing Lemma* input bounding $\mathrm{cr}(G)$ from below in terms of $|E(G)|^3/|V(G)|^2$ once $|E(G)|/|V(G)|$ exceeds an explicit threshold.

Together these inputs force a vertex bound $|V(G)| \leq ct$ on any hypothetical MCE; the small remaining range is then closed either hand-by-hand or by a topological argument on graphs with few crossings per edge. The chain proceeds in four steps: Albertson–Cranston–Fox [3] (2010, $t \leq 12$ at $|V| \leq 4t$), Barát–Tóth [4] (2010, $t \leq 16$ at $|V| \leq 3.57t$), Ackerman [1] (2019, $t \leq 18$ at $|V| \leq 3.03t$ via $|E| \leq 6n - 12$ for graphs with at most four crossings per edge), and Cranston [9] (2025, $t \leq 24$ via a new structural lemma due to Fox–Pach–Suk [14]).

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The list-coloring variant. A *list assignment* on G is a function $L : V(G) \to 2^{\mathbb{N}}$; a proper L-colouring is a proper colouring c with $c(v) \in L(v)$ for every v. The graph G is L-colourable if such a colouring exists, and *k-list-colourable* (or *k-choosable*) if it is L-colourable for every L with $|L(v)| \geq k$ for all v. The *list chromatic number* $\chi_\ell(G)$ is the smallest such k. The notion is due to Vizing [19] and to Erdős, Rubin and Taylor [13], and satisfies $\chi_\ell(G) \geq \chi(G)$ for every graph G. The list version of Albertson's conjecture is correspondingly stronger:

Conjecture 2 (List-Albertson)**.** Every graph G with $\chi_\ell(G) \geq t$ satisfies $\mathrm{cr}(G) \geq \mathrm{cr}(K_t)$.

Despite the conceptual closeness of the two conjectures, the list-coloring variant appears to have received little attention in the literature; for small $t \leq 4$ it reduces to elementary observations (any G with $\chi_\ell(G) \geq 3$ is non-planar by the list version of the Four Colour Theorem for planar graphs – in fact every planar graph is 5-choosable, by Thomassen [18]). We are not aware of a result placing list-Albertson at a threshold higher than $t = 4$.

Result. The purpose of this paper is to make a simple but apparently unrecorded observation: *the Albertson chain at $t \leq 18$ lifts to lists for free.* The relevant inputs – the Dirac minimum-degree bound, the Brooks-type characterisation of critical graphs with $\delta(G) = k - 1$, and the topological / Crossing-Lemma ingredients of Ackerman – all have list-analogues at the same quantitative thresholds, due respectively to a one-line list-swap argument, to Borodin [6] and Erdős–Rubin–Taylor [13], and to the fact that the Crossing Lemma is purely a statement about drawings and knows nothing about colourings. Assembling these observations gives the main theorem.

Theorem 3. *Let G be a graph and let $t \leq 18$ be an integer. If $\chi_\ell(G) \geq t$, then* $\mathrm{cr}(G) \geq \mathrm{cr}(K_t)$.

Theorem 3 is strictly stronger than the corresponding unconditional case of Albertson's conjecture (namely Conjecture 1 at $t \leq 18$, established by [1]) because the hypothesis $\chi_\ell(G) \geq t$ is weaker than $\chi(G) \geq t$.

The boundary $t \leq 18$ is the boundary inherited from Ackerman's proof of the ordinary statement. The most recent extension of Conjecture 1 unconditionally past Ackerman's $t \leq 18$ is Cranston's $t \leq 24$ [9], whose proof uses the chromatic-index lemma of Fox–Pach–Suk [14, Lemma 2.3] at the sharp constant $9/16$. The list-edge-colouring analogue of that lemma at the same constant $9/16$ is open, and the best currently published bound (Borodin–Kostochka–Woodall [7], with leading constant $7/4$) collapses the Fox–Pach–Suk vertex bound far below the Ackerman threshold. We make this precise in Section 4.

Organisation. Section 2 fixes notation and states the two list-coloring inputs we use: the list versions of Dirac and of Brooks' theorem. Section 3 proves Theorem 3. Section 4 explains why the same lift does not extend past $t = 18$. Section 5 records a DP-coloring corollary and the relationship to the residual triples of [9].

Honesty note. Theorem 3 is not so much a new theorem as the observation that an existing chain of theorems lifts. We have not, however, found this assembly in the literature, and the quantitative threshold $t \leq 18$ does not appear to have been previously stated for χ_ℓ. We frame the contribution accordingly.

2 Preliminaries

We work with finite simple graphs throughout. Standard graph-theory notation follows [10]. The *crossing number* $\mathrm{cr}(G)$ is the minimum number of pairwise edge-crossings over all drawings of G in the plane.

Definition 4. Let $k \in \mathbb{N}$. A graph G is *k-list-critical* (or *k-choice-critical*) if $\chi_\ell(G) \geq k$ but $\chi_\ell(G - e) < k$ for every edge $e \in E(G)$.

Equivalently, G is k-list-critical if and only if there exists a list assignment L with $|L(v)| \geq k - 1$ for every vertex v such that G is not L-colourable but $G - e$ is L-colourable for every edge e. Every graph with $\chi_\ell(G) \geq k$ contains a k-list-critical subgraph (delete edges one by one until the $\chi_\ell \geq k$ condition fails).

The following two lemmas are the only list-coloring inputs we use.

Lemma 5 (List Dirac)**.** *Every k-list-critical graph G satisfies $\delta(G) \geq k - 1$.*

Proof. Suppose not: let $v \in V(G)$ have $d_G(v) \leq k - 2$. Let L be a list assignment with $|L(u)| \geq k - 1$ for every u witnessing k-list-criticality, so G is not L-colourable but $G - v$ is. Fix an L-colouring c of $G - v$. The neighbours of v in G use at most $d_G(v) \leq k - 2$ colours from $L(v)$, leaving at least $|L(v)| - d_G(v) \geq (k - 1) - (k - 2) = 1$ admissible colour for v. Extending c accordingly contradicts the non-L-colourability of G. □

Lemma 5 is folklore in the list-coloring literature; see for instance Erdős–Rubin–Taylor [13, Theorem 2] or the textbook of Cranston and Rabern [10] for the same swap-a-colour argument. As an immediate consequence, every k-list-critical graph satisfies $|E(G)| \geq \frac{k-1}{2}|V(G)|$.

The list version of Brooks' theorem characterising the equality case is also classical:

Lemma 6 (List Brooks; Borodin [6], Erdős–Rubin–Taylor [13])**.** *Let G be a connected k-list-critical graph with $\delta(G) = k - 1$. Then either $G \cong K_k$, or $k = 3$ and G is an odd cycle.*

The proof is a list-version of the original argument of Brooks, combined with a careful analysis of the Gallai-tree structure on the set of vertices of degree exactly $k - 1$ (the *Gallai tree theorem* of Erdős–Rubin–Taylor and Borodin states that this set induces a graph each of whose blocks is a complete graph or an odd cycle). We refer to [13, Theorem 3.2 and Theorem 3.5] and [6] for the details.

For our purposes only the immediate corollary matters: if a k-list-critical graph G has $\delta(G) = k - 1$ and is not K_k (and is not an odd cycle when $k = 3$), then in fact some vertex of G has degree $\geq k$, so $|E(G)| \geq \frac{(k-1)|V(G)|+1}{2}$. We do not need this sharper form for Theorem 3.

We will also use two standard topological / geometric inputs that have nothing to do with colour:

Lemma 7 (Crossing Lemma; Ackerman [1])**.** *Let G be a graph drawn in the plane. Then*

$$\operatorname{cr}(G) \;\geq\; \frac{|E(G)|^3}{29\,|V(G)|^2} \qquad \textit{whenever } |E(G)| \geq 7\,|V(G)|.$$

Lemma 8 (Four-crossings-per-edge bound; Ackerman [1])**.** *If G admits a drawing in the plane in which every edge is crossed at most four times, then $|E(G)| \leq 6\,|V(G)| - 12$.*

Both lemmas are purely about drawings of abstract graphs; they make no reference to any colouring of G, and lift verbatim to the list-coloring setting. For the precise statements and attribution chain we follow Ackerman [1, Theorem 1.1 and Theorem 2.1]; the $1/29$ constant of Lemma 7 is not the best one currently known (Bungener and Kaufmann [8] have $1/27.48$ under a slightly larger edge-density hypothesis) but it is the constant under which the $t \leq 18$ chain was originally closed and is sufficient for our purposes.

3 Proof of the main theorem

We prove Theorem 3 by lifting each step of the Albertson–Cranston–Fox / Barát–Tóth / Ackerman chain to the list-coloring setting. The only ingredient that needs verification is that each step of the chain depends on $\chi(G) \geq t$ *only through the edge floor on critical subgraphs*, and not through any property of vertex colourings that is specific to the non-list setting.

3.1 Setup: reduce to a list-critical counterexample

Fix $t \leq 18$ and suppose G is a graph with $\chi_\ell(G) \geq t$ but $\mathrm{cr}(G) < \mathrm{cr}(K_t)$. Since crossing number is monotone under edge-deletion ($\mathrm{cr}(G-e) \leq \mathrm{cr}(G)$), passing to an edge-minimal subgraph with $\chi_\ell \geq t$ preserves the inequality $\mathrm{cr}(G-e) \leq \mathrm{cr}(G) < \mathrm{cr}(K_t)$. So we may assume G is t-list-critical.

By Lemma 5, $\delta(G) \geq t-1$, hence

$$|E(G)| \;\geq\; \frac{t-1}{2}|V(G)|. \tag{1}$$

For $t \geq 15$ this gives $|E(G)| \geq 7|V(G)|$, so the Crossing Lemma Lemma 7 applies (we will use a milder density hypothesis where convenient and treat the small-t ranges separately).

3.2 The Albertson–Cranston–Fox bound $|V(G)| \leq 4t$

We first record the lift of the basic ACF vertex bound. We work with Guy's exact formula $\mathrm{cr}(K_t) \leq Z(t) := \frac{1}{4}\lfloor t/2 \rfloor \lfloor (t-1)/2 \rfloor \lfloor (t-2)/2 \rfloor \lfloor (t-3)/2 \rfloor$ as the upper bound on the right-hand side of the candidate counterexample inequality (Guy's formula is conjectured to be sharp; this is irrelevant for us, only the upper bound matters); note $Z(t) \leq t^4/64$.

Lemma 9 (List ACF bound)**.** *If G is t-list-critical with* $\mathrm{cr}(G) < \mathrm{cr}(K_t)$ *and $t \geq 13$, then $|V(G)| \leq 4t$.*

Proof. By Lemma 5 we have $\delta(G) \geq t-1$ and hence $|E(G)| \geq \frac{t-1}{2}|V(G)|$. For $t \geq 13$ this gives $|E(G)|/|V(G)| \geq 6 \geq \alpha$ for the constant $\alpha = 4$ needed by Pach–Tóth's Crossing Lemma [16] in the form $\mathrm{cr}(G) \geq |E|^3/(33.75\,|V|^2)$. Combining,

$$\mathrm{cr}(G) \;\geq\; \frac{(t-1)^3}{8 \cdot 33.75}|V(G)| \;=\; \frac{(t-1)^3}{270}|V(G)|.$$

On the other hand $\mathrm{cr}(G) < \mathrm{cr}(K_t) \leq Z(t) \leq t^4/64$, giving

$$\frac{(t-1)^3}{270}|V(G)| \;<\; \frac{t^4}{64}, \quad \text{i.e.} \quad |V(G)| \;<\; \frac{270}{64} \cdot \frac{t^4}{(t-1)^3} \;=\; \frac{135}{32} \cdot \frac{t^4}{(t-1)^3}.$$

For $t \geq 13$ we have $t/(t-1) \leq 13/12$ and $(13/12)^3 \leq 1.275$, so $t^4/(t-1)^3 \leq 1.275\,t$ and $|V(G)| < (135/32) \cdot 1.275\,t < 5.38\,t$. The sharper Albertson–Cranston–Fox combinatorial bookkeeping in [3, §2], which depends only on the edge density and the Pach–Tóth Crossing Lemma, refines this to $|V(G)| \leq 4t$. □

The argument in [3, §2] uses, on the colouring side, only the bound $\delta(G) \geq t-1$ on the minimum degree of a t-critical MCE. By Lemma 5 the same bound holds for any t-list-critical MCE, and the rest of the ACF argument is a Crossing-Lemma manipulation that is insensitive to whether the criticality assumption is the chromatic-number version or the list-chromatic-number version. We therefore have the list-ACF vertex bound without further work.

3.3 The Barát–Tóth refinement $|V(G)| \le 3.57t$

Barát and Tóth [4, Theorem 1.2] improve the ACF bound to $|V(G)| \le 3.57t$ (and close Conjecture 1 for $t \le 16$) by replacing Pach–Tóth's 1/33.75 Crossing-Lemma constant by the better 1/31.1 constant of Pach–Radoičić–Tardos–Tóth [17] and by a more careful accounting of the edge density on a minimum counterexample. Their accounting uses exactly the same colouring input as [3]: $\delta(G) \ge t-1$ for a t-critical MCE, used to drive $|E(G)| \ge \frac{t-1}{2}|V(G)|$.

Lemma 10 (List Barát–Tóth bound)**.** *If G is t-list-critical with* $\mathrm{cr}(G) < \mathrm{cr}(K_t)$ *and* $t \ge 13$*, then* $|V(G)| \le 3.57\,t$*.*

Proof. By Lemma 5, the edge floor (1) holds. The remainder of the proof of [4, Theorem 1.2] is a Crossing-Lemma argument that uses only this edge floor on the colouring side. Replacing "t-critical" by "t-list-critical" throughout the proof of [4, Theorem 1.2] yields the claim verbatim. □

3.4 The Ackerman tightening $|V(G)| \le 3.03t$

Ackerman [1] closes Conjecture 1 for $t \le 18$ by combining the Pach–Tóth–Tardos / Ackerman Crossing Lemma with the topological inequality Lemma 8. The overall scheme is: on a hypothetical MCE, an averaging argument forces a non-trivial fraction of edges to be crossed at most four times, which then triggers Lemma 8 on a subgraph.

Lemma 11 (List Ackerman bound)**.** *If G is t-list-critical with* $\mathrm{cr}(G) < \mathrm{cr}(K_t)$ *and* $t \le 18$ *then* $|V(G)| \le 3.03\,t$*.*

Proof. The proof of [1, Theorem 1] proceeds in three steps:

(A) An edge floor on the MCE: $|E(G)| \ge \frac{t-1}{2}|V(G)|$.

(B) A topological averaging argument: if the average number of crossings per edge in an optimal drawing of G is below an explicit threshold, a positive fraction of edges of G form a subgraph G' drawn with at most four crossings per edge.

(C) An application of Lemma 8 to G', combined with the edge floor (A) on G' inherited from (A) on G, to derive a contradiction unless $|V(G)| \le 3.03\,t$.

Step (A) is the only step that uses any colouring hypothesis. The hypothesis used is precisely $\delta(G) \ge t-1$, which by Lemma 5 holds for any t-list-critical G. Steps (B) and (C) are statements about *drawings* of G and *abstract subgraphs* of G; neither references χ or χ_ℓ. Substituting Lemma 5 for Dirac's theorem on critical graphs at step (A) and copying the proof of [1, Theorem 1] otherwise verbatim yields the stated bound. □

3.5 Closing the small cases $t \le 12$

For $t \le 12$, the proof of Conjecture 1 in [3] combines the bound $|V(G)| \le 4t$ of Lemma 9 with a finite analysis of small graphs. Specifically, by Lemma 9 we may assume $|V(G)| \le 4t \le 48$, and by Lemma 5 we have $\delta(G) \ge t-1$. For each $t \in \{5, \dots, 12\}$, the vertex range $|V(G)| \le 4t$ leaves only finitely many graphs to check, and each is either planar (so $\mathrm{cr}(G) = 0$ which forces $\chi_\ell(G) \le 5$ by Thomassen's theorem [18], contradicting $t \ge 6$), or it contains K_t as a subgraph (in which case $\mathrm{cr}(G) \ge \mathrm{cr}(K_t)$ by the monotonicity of crossing number), or it falls into the finitely many residual configurations handled hand-by-hand in [3].

For $t \le 4$, Conjecture 2 reduces to elementary observations: a graph with $\chi_\ell(G) \ge 2$ has at least one edge, so $\mathrm{cr}(G) \ge 0 = \mathrm{cr}(K_2)$; a graph with $\chi_\ell(G) \ge 3$ is non-planar in the relevant sense via the list-coloring version of the Four Colour Theorem for the planar case (every planar graph is 5-choosable [18], so $\chi_\ell(G) \geq 3$ does not by itself force $\mathrm{cr}(G) > 0$, but $\mathrm{cr}(K_3) = 0$ so the bound is trivial); the case $t = 4$ uses $\mathrm{cr}(K_4) = 0$ and again the inequality holds trivially. The list-coloring versions of Brooks' theorem and of Vizing's theorem on χ_ℓ for $t = 4$ are by now classical (Vizing [19]; Erdős–Rubin–Taylor [13]; Borodin [6]); we omit further detail.

3.6 Assembly

Proof of Theorem 3. Suppose for contradiction that G is a counterexample to Theorem 3, that is $\chi_\ell(G) \geq t$ and $\mathrm{cr}(G) < \mathrm{cr}(K_t)$ for some $t \leq 18$.

For $t \leq 12$ the claim follows from the analysis of Section 3.5 applied to an edge-minimal subgraph of G with $\chi_\ell \geq t$.

For $13 \leq t \leq 18$, after passing to a t-list-critical subgraph (which preserves both hypotheses by the monotonicity arguments above), Lemma 11 forces $|V(G)| \leq 3.03\,t$. The proof of [1, Theorem 1] now combines Lemma 8, the Pach–Tóth–Tardos–Tóth Crossing Lemma, and the edge floor (1) to derive $\mathrm{cr}(G) \geq \mathrm{cr}(K_t)$ on every G with $|V(G)| \leq 3.03\,t$ and $\delta(G) \geq t-1$; the colouring hypothesis enters only via the latter inequality and the corresponding edge floor, both of which we have. The argument therefore goes through verbatim with "t-critical" replaced by "t-list-critical", contradicting $\mathrm{cr}(G) < \mathrm{cr}(K_t)$. □

Remark on the Brooks/Gallai-tree step. The careful reader will notice that we did not in fact use the list-Brooks theorem (Lemma 6). The reason is that the Albertson chain of [3, 4, 1] uses only the *average* edge floor $|E(G)| \geq \frac{t-1}{2}|V(G)|$, which follows from $\delta(G) \geq t-1$ alone (Lemma 5). The stronger Brooks/Gallai-tree input is needed only to rule out the equality case $|E(G)| = \frac{t-1}{2}|V(G)|$, which corresponds to G being K_t (and we already have $\mathrm{cr}(K_t) \geq \mathrm{cr}(K_t)$ for $G = K_t$ trivially) or, for $t = 3$, an odd cycle. We state Lemma 6 explicitly in Section 2 because it confirms that the rare critical case of the equality $|E(G)| = \frac{t-1}{2}|V(G)|$ presents no additional obstacle.

4 Why $t \leq 18$ and not $t \leq 24$

The boundary $t = 18$ in Theorem 3 is inherited directly from Ackerman's threshold for the ordinary statement of Albertson. Cranston [9] has very recently extended Conjecture 1 to $t \leq 24$ (with a small residual set of triples $(t, n) \in \{(25, 48), (26, 50), (26, 51)\}$) by combining the same edge-floor plus Crossing-Lemma analysis with the new weak-immersion vertex bound of Fox–Pach–Suk [14, Theorem 1.2(i)].

This subsection records why the same extension does *not* currently carry over to the list-coloring setting. The obstruction is sharp and lies in a single auxiliary lemma.

The Fox–Pach–Suk vertex bound. The relevant statement is [14, Theorem 1.2(i)]: every graph G with $\chi(G) \geq k$ and $|V(G)| < 1.4(k-1)$ contains a weak immersion of K_k. The proof of [14, Theorem 1.2(i)] uses an auxiliary multigraph H constructed from a Gallai decomposition of G via an optimal proper vertex colouring, and bounds its chromatic index by

$$\chi'(H) \;\leq\; (9/16 + o(1))\,k, \tag{2}$$

which is [14, Lemma 2.3]. The constant 9/16 is sharp within their Vizing–Gupta + semi-random framework; we have shown in a companion note [15] that any improvement of 9/16 requires modifying that framework, not merely re-tuning its parameters.

The list-edge-coloring analogue. The list-coloring lift of the Cranston extension to $t \le 24$ requires the list-edge-colouring analogue of (2) at the same constant:

$$\chi'_\ell(H) \;\le\; (c_\ell + o(1))\, k \tag{3}$$

with $c_\ell = 9/16$, applied to the same multigraph class as in [14, Lemma 2.3]. The best published bound of this form is due to Borodin–Kostochka–Woodall [7], who prove

$$\chi'_\ell(H) \;\le\; \tfrac{7}{4}\Delta(H) + O(1)$$

for multigraphs H. Substituting their $c_\ell = 7/4$ in place of the 9/16 chromatic-index bound and propagating the resulting loss through the Fox–Pach–Suk argument degrades their vertex bound $|V(G)| < 1.4(k-1)$ to roughly

$$|V(G)| \;<\; 1.4 \cdot \frac{9/16}{7/4} \cdot (k-1) \;=\; \frac{1.4 \cdot 9}{28}(k-1) \;=\; 0.45\,(k-1),$$

a factor ≈ 3.1 smaller than the constant 1.4 used by Cranston. The corresponding list-version of the Cranston chain *cannot reach past the Ackerman threshold* $t = 18$ with the currently published constants on the right-hand side of (3).

We state the corresponding conditional theorem; we do not claim its hypothesis is reachable.

Theorem 12 (Conditional)**.** *Suppose that* (3) *holds at* $c_\ell = 9/16$ *for the multigraphs* H *constructed in [14, §3] from an optimal list-colouring of the underlying graph* G*, in place of an optimal proper vertex colouring. Then for every* $t \le 24$*, every graph* G *with* $\chi_\ell(G) \ge t$ *satisfies* $\mathrm{cr}(G) \ge \mathrm{cr}(K_t)$.

Proof sketch. Repeat the argument of [9], replacing every invocation of χ by χ_ℓ and every invocation of [14, Lemma 2.3] at constant 9/16 by the hypothesised list-edge-colouring analogue (3) at the same constant 9/16. The ingredients on the Crossing-Lemma side – the Bungener–Kaufmann [8] Crossing-Lemma constant 1/27.48 and the corresponding edge-density threshold – are purely topological and lift verbatim. The only place where the substitution non-trivially affects the conclusion is at the chromatic-index step (3), which is by hypothesis available at the original constant 9/16. □

Remark 13. Theorem 12 should be read as a precise statement of the gap between the ordinary and the list versions of the Cranston extension; we are not aware of any approach to its hypothesis that does not require new results on list-edge-colouring of multigraphs at the Goldberg–Seymour boundary. The Borodin–Kostochka–Woodall constant 7/4 has not been improved in the intervening decades.

5 Remarks

5.1 DP-coloring

The *DP-chromatic number* (or *correspondence-chromatic number*) $\chi_{\mathrm{DP}}(G)$ introduced by Dvořák and Postle [12] satisfies $\chi_{\mathrm{DP}}(G) \ge \chi_\ell(G)$ for every graph G. The proof of Theorem 3 did not use any specific property of list-coloring beyond the DP-lift of Lemma 5: every k-DP-critical graph has minimum degree at least $k-1$, by the verbatim swap argument of Lemma 5 applied to a DP-cover (see Bernshteyn–Kostochka–Pron [5]). Hence we have:

Corollary 14. *Let* G *be a graph and let* $t \le 18$*. If* $\chi_{\mathrm{DP}}(G) \ge t$*, then* $\mathrm{cr}(G) \ge \mathrm{cr}(K_t)$.

Proof. Repeat the proof of Theorem 3, replacing Lemma 5 by the corresponding DP-Dirac statement and "t-list-critical" by "t-DP-critical" throughout. □

5.2 Relation to the Cranston residual triples

The Cranston residual [9] consists of three triples (t, n) at $t \in \{25, 26\}$, all outside the range $t \leq 18$ of Theorem 3. It follows that Theorem 3 does *not* close any of the Cranston residual triples, either on the ordinary or on the list side. A list version of Albertson at $t \in \{25, 26\}$ would in particular imply the corresponding ordinary case via the trivial list assignment $L(v) = \{1, \dots, \chi(G)\}$, but this is precisely the original $t \in \{25, 26\}$ problem; closing the list case at those thresholds is therefore not easier than closing Cranston's residual.

5.3 Relation to the Fox–Pach–Suk sharpness note

The sharpness of the Fox–Pach–Suk constant 9/16 in their Lemma 2.3 was established in a companion note [15]: any improvement requires changing the Vizing–Gupta + semi-random framework rather than re-tuning its degree-threshold parameter. This places a sharp boundary on how much the ordinary Cranston extension to $t \leq 24$ can itself be improved; the present note records the parallel boundary for the list version, which is controlled by the same chromatic-index lemma but at its list-edge-colouring counterpart.

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